

Signal Journey: Sense of Connection in a Two-Player VR Game Through a Proximity-based Interactive Experience

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Figure 1: VR game scenes for long-range or mid-range, close-range, and very close-range

Abstract

This project aims to build a sense of connection in a remote two-player VR game by offering proximity-based interaction with audio and video connectivity. We develop a co-dependent game system where virtual proximity, or the distance between two players are assumed to reflect emotional closeness of the two individuals. The VR metaphorically reflect and promote players' mutual engagement. Positioned at opposite ends of the virtual environment, the two players exchange audio and visual signals to locate each other until they come close. As they get nearer, visual and audio elements, such as color saturation and diverse sounds, are designed to enhance their connectivity, reflecting their emotional closeness and awareness of each other's presence. After gameplay, the players' locations are recorded and displayed on a map.

CCS Concepts

• Applied computing → Media arts; • Human-centered computing → Virtual reality; • Software and its engineering → Interactive games.

Keywords

Co-presence, Proximity, Virtual Reality Game, Sensory Experience

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1 Introduction and Background

Proximity in VR refers to the distance between two players, which can vary depending on their relationship [3]. Therefore, proximity is a measure of not just technical, but also emotional closeness [2]. In many collaborative games, players can form shared memories and build relationships through interactive storytelling and gameplay mechanics [1]. In our project “Signal Journey,” we explore proximity to determine emotional closeness between players, enhancing their awareness that they are actually forming a connection with one another. The concept of proximity is realized as an immersive sensory journey to connect with one another in VR.

2 Design and Implementation

The emotional implications of distance and proximity between two players in VR are conveyed through the narrative of cool air and warm air interacting with one another. The two players, positioned at opposite ends of the whimsical world, start off on a journey to find each other, relying on sensory signals. The cool air is represented by a blue shape, and the warm air is represented by a red shape. The visual and audio systems of the game progress through four stages of distance: long-range, mid-range, close-range, and very close-range. As the players draw closer, the color, visibility, and directional cues intensify, providing more information about the

Table 1: Four stages of progression and changes in audio and visual game systems

	Long-Range	Mid-Range	Close-Range	Very Close-Range
Environmental Color	Low saturation	Default color	High saturation, color inversion	Default color
Visibility of the Corresponding Player	Invisible	Invisible	Partially visible through trails only	Visible
Walking Speed	Normal	Normal	Slower	Slower
First Player's Sound	High	Low	No sound	Mutual aural harmony
Atmospheric Sound	Normal	Low	None	None
Corresponding Player's Signature Tune	Audible when receiving Flower Petals	Audible when receiving Flower Petals	Audio gets clearer when Players get closer	High
Directional Cue	Direction of the flower petal	Direction of the flower petal	Beeping sound guide	None
Signals Between Players	Send flower petals	Send flower petals	Find the angle	Unite

corresponding player. When the players unite, the tale of cool air and warm air concludes with the metaphor of a mythical rain and a purple cloud, symbolizing their coming together. Upon completing the game, traces of the players' journey are recorded and displayed on a final map. This map serves as a meaningful reflection of their shared experience.

Sound plays a crucial role as a distinctive signal, conveying information and enabling the two players to navigate toward each other. The sound system includes four distinct elements: the signature tunes of the two players, the atmospheric sound of the environment, sound effects for events, and sonic guides. All sound effects were composed using notes in the same key (B major) to ensure they blend together naturally. The spatial sound was created using the Spatial Audio feature in the Unity engine. The parameters for reverb and the low-pass filter were mapped according to the positions of the two players to reflect the sense of distance.

2.1 The Gameplay

At the long-range stage, the first player sends flower petals and watches them float toward the corresponding player to determine the direction to proceed. When the corresponding player receives the flower petals, they can hear the first player's signature tune. Feedback for the first player's movement, which was prominent at long range, gradually diminishes, making the presence and interaction of the corresponding player more prominent at close range. To navigate in close-range, the players focus on the direction from which each other's sounds are coming. At the very close-range stage, the footstep-like sound of the first player becomes inaudible, while the other player's signature tune grows louder, adding another layer of sound. At this point, the players slow down to extend the moment they can spend together. At very close range, they finally become co-present and can see each other. After the game, the players receive a summary that highlights the key moments and paths of their journey.

2.2 System Configuration

Remote connections for multiplayer functionality are established using PUN2 (Photon Unity Networking), allowing two remote players to connect to the Photon network via an internet connection. Each client synchronizes the positions of the players and exchanges information in real time through the internal functions provided by PUN2. The virtual reality experience is designed to run on Quest 2 with two controllers. Using controller input and haptic feedback, locomotion is achieved through an arm-swinging motion. Swinging one's hand while holding the virtual flower using the controller trigger causes the flower to be sent.

3 CONCLUSION

The "Signal Journey" project contributes to VR by demonstrating how proximity-based interactions can effectively foster emotional connections between remote players. It emphasizes the dual role of proximity as both a technical and emotional metric in VR, highlighting its potential for enhancing interpersonal connections in future games and applications. The project incorporated visual motifs from emotive gaming contexts and dynamically implemented sensory cues to guide players toward meaningful interactions in the virtual environment.

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